

2. FraudTransactionDetection (Linear Search)

```
import java.util.*;

Public class FraudTransactionDetection
{
    Public static boolean isFraudTransaction(int transaction[],
                                             int suspicious)
    {
        for (int i=0; i<transaction.length; i++)
        {
            if (transaction[i] == suspicious)
            {
                return true;
            }
        }
        return false;
    }

    Public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int transaction[] = new int[n];
        for (int i=0; i<n; i++)
        {
            transaction[i] = sc.nextInt();
        }

        int suspicious = sc.nextInt();

        if (isFraudTransaction(transaction, suspicious))
        {
            System.out.println("Fraud Transaction Found");
        }
        else
        {
            System.out.println("Transaction not found");
        }
    }
}
```

```
20
21 public static void main(String[] args) {
22     System.out.print("length of array=");
23
24
25     Scanner sc= new Scanner(System.in);
26     int n=sc.nextInt();
27
28     int transaction []= new int [n];
29
30     for(int i=0; i<n;i++)
31     {
32         transaction[i]=sc.nextInt();
33     }
34
35     System.out.print("enter the suspicious id ");
36     int suspicious= sc.nextInt();
37
38     if (isFraudTransaction(transaction, suspicious)) {
39         System.out.println("Fraud Transaction Found");
40     } else {
41         System.out.println("Transaction Not Found");
42     }
43 }
44
45 }
46
```

Console X

<terminated> FraudTransactionDetection [Java Application] C:\Users\Suraj Yadav\p2\pool\plugins\org.e

length of array=3

23

45

44

enter the suspicious id 45

Fraud Transaction Found

```

import java.util.*;
Public class SeverLogSearch
{
    Public static boolean islogid(int logID[], int TargetID)
    {
        int start = 0, end = logID.length - 1;
        while (start <= end)
        {
            int mid = (start + end) / 2;
            if (logID[mid] == TargetID)
            {
                return true;
            }
            else if (logID[mid] < TargetID)
            {
                start = mid + 1;
            }
            else
            {
                end = mid - 1;
            }
        }
        return false;
    }

    Public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int logID[] = new int[n];
        for (int i = 0; i < n; i++)
        {
            logID[i] = sc.nextInt();
        }
        Arrays.sort(logID);
        int TargetID = sc.nextInt();
        if (islogid(logID, TargetID))
        {
            System.out.println("log found");
        }
    }
}

```

```

2 import java.util.*;
3 public class SeverLogSearch {
4
5     public static boolean islogid(int logID[], int TargetlogID)
6     {
7         int start=0;
8         int end=logID.length-1;
9
10        while(start<=end)
11        {
12            int mid=(start+end)/2;
13            if(logID[mid]==TargetlogID)
14            {
15                return true;
16            }
17            else if(logID[mid]<TargetlogID)
18            {
19                start=mid+1;
20            }
21            else
22            {
23                end=mid-1;
24            }
25        }
26        return false;
27    }
28
29    public static void main(String[] args) {
30        // TODO Auto-generated method stub
31        Scanner sc= new Scanner(System.in);
32        System.out.print("Enter the number of log in ID= ");
33        int n= sc.nextInt();
34
35        int logID[]= new int[n];
36
37        System.out.print("Enter the log in ID ");
38        for(int i=0 ; i<n; i++)
39        {
40            logID[i]=sc.nextInt();
41        }
42    }
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```

Console X

<terminated> SeverLogSearch [Java Application] C:\Users\Suraj Yadav\

```

Enter the number of log in ID= 4
Enter the log in ID 45
56
78
79
Enter the target Element 56
log Found

```

2. Sort Employee (Bubble Sort)

```
import java.util.Scanner;

public class Main {

    public static void bubbleSort(int salary[], int n)
    {
        for (int i = 0; i < n - 1; i++)
        {
            for (int j = 0; j < n - i - 1; j++)
            {
                if (salary[j] > salary[j + 1])
                {
                    salary[j]
                    int temp = salary[j];
                    salary[j] = salary[j + 1];
                    salary[j + 1] = temp;
                }
            }
        }
    }

    public static void main (String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the number of Employee");
        int n = sc.nextInt();
        int salary[] = new int[n];
        for (int i = 0; i < n; i++)
        {
            salary[i] = sc.nextInt();
        }
        bubbleSort(salary, n);
        for (int i = 0; i < n; i++)
        {
            System.out.println(salary[i] + " ");
        }
    }
}
```

```
1 package ARRAY;
2 import java.util.Scanner;
3 public class Main {
4     static void bubbleSort(int[] salary, int n) {
5         for (int i = 0; i < n - 1; i++) {
6             for (int j = 0; j < n - i - 1; j++) {
7                 if (salary[j] > salary[j + 1]) {
8                     int temp = salary[j];
9                     salary[j] = salary[j + 1];
10                    salary[j + 1] = temp;
11                }
12            }
13        }
14    }
15
16    public static void main(String[] args) {
17        Scanner sc = new Scanner(System.in);
18        System.out.print("Enter the number of Employee=");
19        int n = sc.nextInt();
20        int[] salary = new int[n];
21        System.out.println("Enter the salaries of employee=");
22        for (int i = 0; i < n; i++) {
23            salary[i] = sc.nextInt();
24        }
25
26        bubbleSort(salary, n);
27
28        for (int i = 0; i < n; i++) {
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```

Console X

```
<terminated> Main [Java Application] C:\Users\Suraj Yadav\p2\pool\plugins\org.eclipse.justi.openjdkhotspot1
Enter the number of Employee=5
Enter the salaries of employee=
34
4500
56000
44
34567
34 44 4500 34567 56000
```



```

4. Sort website ResponseTime (Insertion Sort)
import java.util.*;

Public class WebsiteResponseTimeSort {
    Public static void insertionSort(int arr[])
    {
        for (int i = 1; i < arr.length; i++)
        {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key)
            {
                arr[j + 1] = arr[j];
                j = j - 1;
            }
            arr[j + 1] = key;
        }
    }

    Public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        insertionSort(arr);
        for (int i = 0; i < n; i++)
        {
            System.out.print(arr[i] + " ");
        }
    }
}

```

```

2 import java.util.*;
3
4 public class WebsiteResponseTimeSort {
5     public static void insertionSort(int arr[]) {
6         for (int i = 1; i < arr.length; i++) {
7             int key = arr[i];
8             int j = i - 1;
9
10            while (j >= 0 && arr[j] > key) {
11                arr[j + 1] = arr[j];
12                j = j - 1;
13            }
14            arr[j + 1] = key;
15        }
16    }
17
18    public static void main(String[] args) {
19        Scanner sc = new Scanner(System.in);
20
21        System.out.print("Enter the size of response time array ");
22        int n = sc.nextInt();
23        int arr[] = new int[n];
24
25        System.out.println("Enter the response time in array ");
26        for (int i = 0; i < n; i++) {
27            arr[i] = sc.nextInt();
28        }
29
30        insertionSort(arr);
31
32        for (int i = 0; i < n; i++) {
33            System.out.print(arr[i] + " ");
34        }
35    }
36 }

```

Console X

<terminated> WebsiteResponseTimeSort [Java Application] C:\Users\Suraj Yadav\p2\pool\N

Enter the size of response time array=3

Enter the response time in array

34

56

32

32 34 56

Prioritize Bug Reports (Selection Sort)

```

import java.util.*;

public class PriorityScore {

    public static void selectionSort(int arr[], int n) {
        for (int i = 0; i < n - 1; i++) {
            int min = i;
            for (int j = i + 1; j < n; j++) {
                if (arr[j] < arr[min]) {
                    min = j;
                }
            }
            int temp = arr[i];
            arr[i] = arr[min];
            arr[min] = temp;
        }
    }

    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        selectionSort(arr, n);
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
    }
}

```

```

3 public class PriorityScore {
4     static void selectionSort(int[] arr, int n) {
5         for (int i = 0; i < n - 1; i++) {
6             int min = i;
7
8             for (int j = i + 1; j < n; j++) {
9                 if (arr[j] < arr[min]) {
10                     min = j;
11                 }
12             }
13             int temp = arr[i];
14             arr[i] = arr[min];
15             arr[min] = temp;
16         }
17     }
18
19     public static void main(String[] args) {
20         Scanner sc = new Scanner(System.in);
21         System.out.print("Enter the number of size of priority scores =");
22         int n = sc.nextInt();
23         int[] arr = new int[n];
24
25         System.out.println("Enter the element in array of priority score");
26         for (int i = 0; i < n; i++) {
27             arr[i] = sc.nextInt();
28         }
29         selectionSort(arr, n);
30
31         for (int i = 0; i < n; i++) {
32             System.out.print(arr[i] + " ");
33         }
34     }
35 }

```

Console X

```

<terminated> PriorityScore [Java Application] C:\Users\Suraj Yadav\p2\pool\plugins\org.eclipse.justj.openjdk
Enter the number of size of priority scores =4
Enter the element in array of priority score
46
23
45
22
22 23 45 46

```